TOC: Group Project.

NFA Design to DFA with subsets.

1. "free"

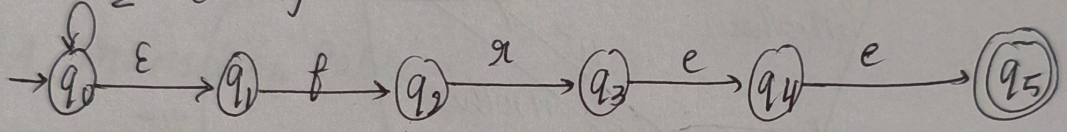
$\{Q, \Sigma, \delta, q_0, F\}$

$\Sigma = \{f, r, e, other\}$

$F = \{q_5\}$

NFA diagram:-

Σ^* (all symbols in Σ)



NFA Transition Table:-

	f	r	e	other	ε
→q_0	$\{q_0\}$	$\{q_0\}$	$\{q_0\}$	$\{q_0\}$	$\{q_1\}$
q_1	$\{q_2\}$	ϕ	ϕ	ϕ	ϕ
q_2	ϕ	$\{q_3\}$	ϕ	ϕ	ϕ
q_3	ϕ	ϕ	$\{q_4\}$	ϕ	ϕ
q_4	ϕ	ϕ	$\{q_5\}$	ϕ	ϕ
*q_5	ϕ	ϕ	ϕ	ϕ	ϕ

$\varepsilon\text{-clos}(q_0) = \{q_0, q_1\}$

$\varepsilon\text{-clos}(q_1) = \{q_1\}$

$\varepsilon\text{-clos}(q_2) = \{q_2\}$

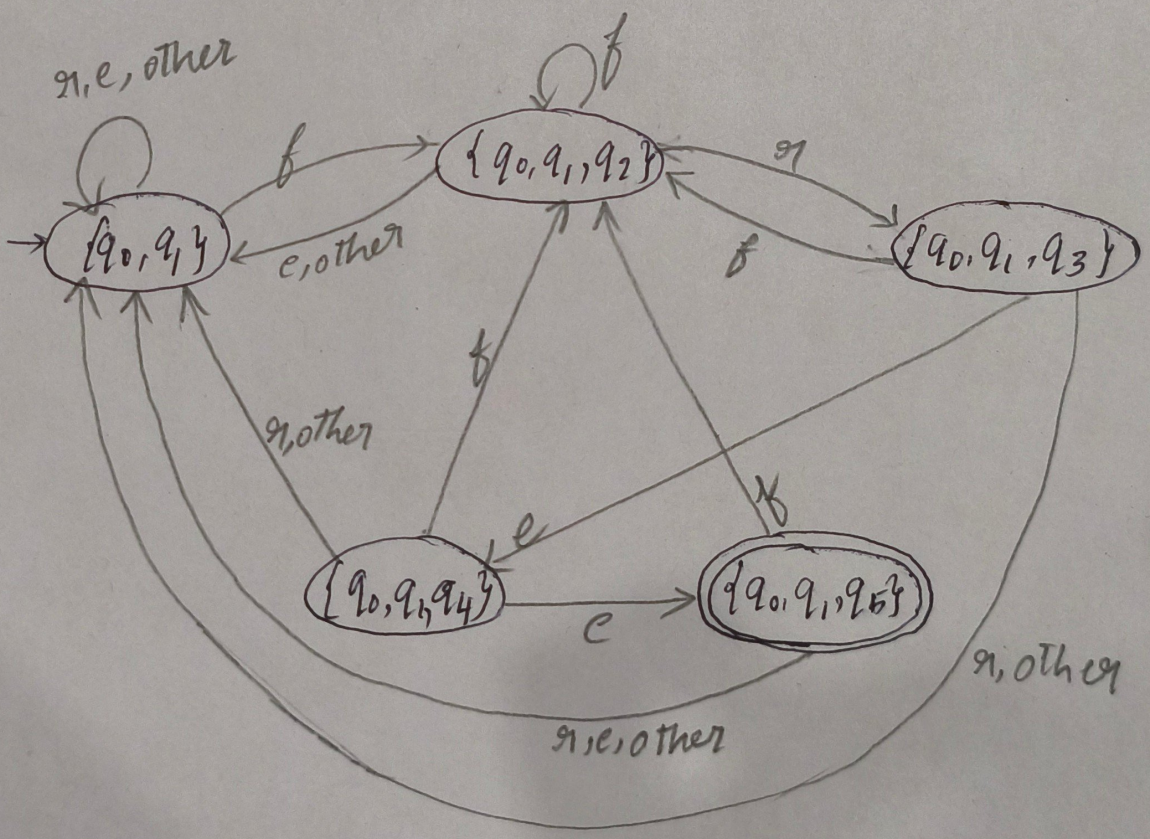
$\varepsilon\text{-clos}(q_3) = \{q_3\}$

$\varepsilon\text{-clos}(q_4) = \{q_4\}$

$\varepsilon\text{-clos}(q_5) = \{q_5\}$

DFA Transition Table; and diagram:-

	f	n	e	other
→ $\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$
$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_3\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$
$\{q_0, q_1, q_3\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1\}$	$\{q_0, q_1, q_4\}$	$\{q_0, q_1\}$
$\{q_0, q_1, q_4\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1\}$	$\{q_0, q_1, q_5\}$	$\{q_0, q_1\}$
$\{q_0, q_1, q_5\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$



2. win.

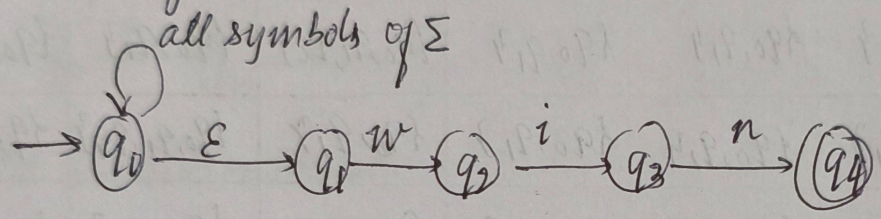
$\{Q, \Sigma, \delta, q_0, F\}$ where,

$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{w, i, n, other\}$

$F = \{q_4\}$

NFA- diagram:-



NFA Transition Table:-

	w	i	n	other	ε
→q_0	$\{q_0\}$	$\{q_0\}$	$\{q_0\}$	$\{q_0\}$	$\{q_1\}$
q_1	$\{q_2\}$	ϕ	ϕ	ϕ	ϕ
q_2	ϕ	$\{q_3\}$	ϕ	ϕ	ϕ
q_3	ϕ	ϕ	$\{q_4\}$	ϕ	ϕ
*q_4	ϕ	ϕ	ϕ	ϕ	ϕ

$\varepsilon_clos(q_0) = \{q_0, q_1\}$

$\varepsilon_clos(q_1) = \{q_1\}$

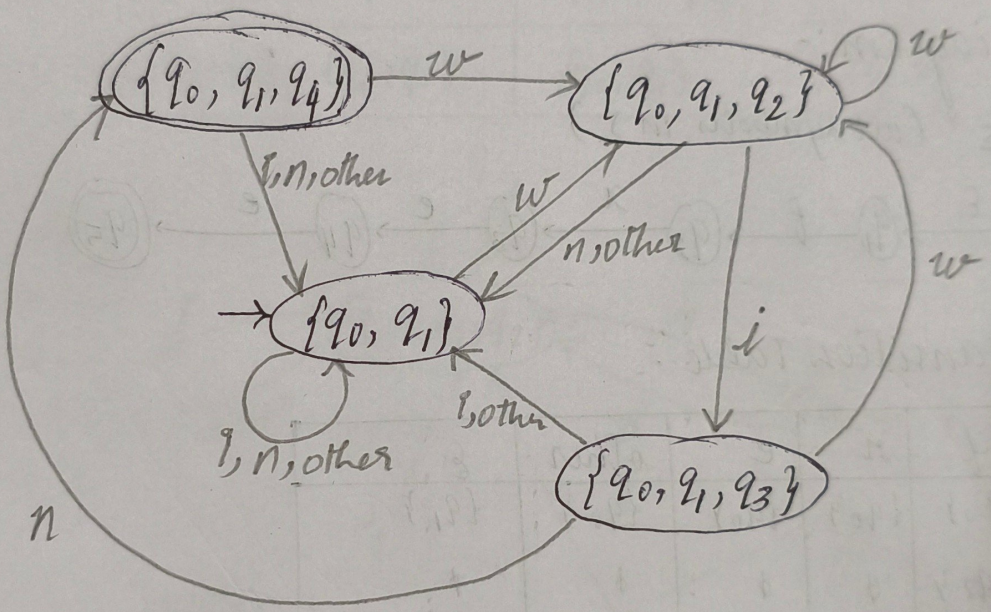
$\varepsilon_clos(q_2) = \{q_2\}$

$\varepsilon_clos(q_3) = \{q_3\}$

$\varepsilon_clos(q_4) = \{q_4\}$

DFA Transition Table and diagram:-

	w	i	n	other
→ {q0, q1}	{q0, q1, q2}	{q0, q1}	{q0, q1}	{q0, q1}
{q0, q1, q2}	{q0, q1, q2}	{q0, q1, q3}	{q0, q1}	{q0, q1}
{q0, q1, q3}	{q0, q1, q2}	{q0, q1}	{q0, q1, q4}	{q0, q1}
* {q0, q1, q4}	{q0, q1, q2}	{q0, q1}	{q0, q1}	{q0, q1}



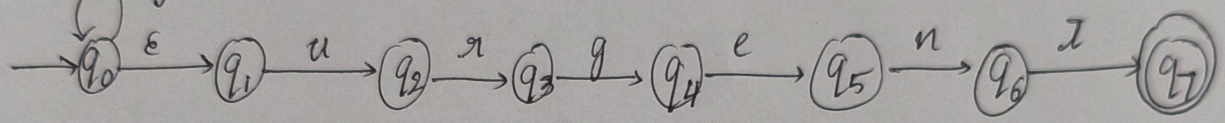
3. Pattern: "urgent"

$\{Q, \Sigma, \delta, q_0, F\}$

$\Sigma = \{u, r, g, e, n, t, other\}$

$F = \{q_7\}$

NFA diagram:-



NFA Transition Table:-

	u	r	g	e	n	t	other	ε
→q_0	$\{q_0\}$	$\{q_0\}$	$\{q_0\}$	$\{q_0\}$	$\{q_0\}$	$\{q_0\}$	$\{q_0\}$	$\{q_1\}$
q_1	$\{q_2\}$	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ
q_2	ϕ	$\{q_3\}$	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ
q_3	ϕ	ϕ	$\{q_4\}$	ϕ	ϕ	ϕ	ϕ	ϕ
q_4	ϕ	ϕ	ϕ	$\{q_5\}$	ϕ	ϕ	ϕ	ϕ
q_5	ϕ	ϕ	ϕ	ϕ	$\{q_6\}$	ϕ	ϕ	ϕ
q_6	ϕ	ϕ	ϕ	ϕ	ϕ	$\{q_7\}$	ϕ	ϕ
q_7	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ	ϕ

$\varepsilon_clos(q_0) = \{q_0, q_1\}$
$\varepsilon_clos(q_1) = \{q_1\}$
$\varepsilon_clos(q_2) = \{q_2\}$
$\varepsilon_clos(q_3) = \{q_3\}$
$\varepsilon_clos(q_4) = \{q_4\}$
$\varepsilon_clos(q_5) = \{q_5\}$
$\varepsilon_clos(q_6) = \{q_6\}$
$\varepsilon_clos(q_7) = \{q_7\}$

DFA Transition Table.

	u	r	g	e	n	t	other
→ $\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$
$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_3\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$
$\{q_0, q_1, q_3\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1\}$	$\{q_0, q_1, q_4\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$
$\{q_0, q_1, q_4\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1, q_5\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$
$\{q_0, q_1, q_5\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1, q_6\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$
$\{q_0, q_1, q_6\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1, q_7\}$	$\{q_0, q_1\}$
* $\{q_0, q_1, q_7\}$	$\{q_0, q_1, q_2\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$	$\{q_0, q_1\}$

